Module 06: Learning in Embodied Cognition — Somatic Learning

Learning Objectives

1. Define **somatic learning** as the updating of the embodied generative model through physical practice, contemplative training, and experiential engagement.
2. Analyze how **practice, repetition, and corrective feedback** implement Dirichlet parameter learning in the somatic domain.
3. Apply the Active Inference framework to explain the **stages of embodied skill acquisition** and the role of plateaus, breakthroughs, and regressions.

Introduction

You cannot learn to swim from a textbook. You cannot learn to cook by reading recipes. You cannot learn to feel empathy by studying psychology. Embodied learning requires *doing* — engaging the body in repeated cycles of prediction, action, and error correction that progressively refine the somatic generative model.

This module examines how the embodied agent learns — not through symbolic instruction alone but through the iterative process of somatic prediction error minimization.

Key Concepts

1. Motor Learning as Parameter Updating

When learning a new physical skill (juggling, typing, a yoga pose), the embodied agent's generative model undergoes systematic refinement:

- **Early learning** (high prediction error): The proprioceptive predictions are rough — the body doesn't yet know where to move. Each attempt generates large prediction errors. Learning rate is high.

- **Intermediate learning** (declining prediction error): The predictions become more accurate. The agent can perform the basic pattern but still generates errors on precision demands (timing, force calibration). Learning rate is moderate.
- **Late learning** (minimal prediction error): The predictions are precise and habitual. The policy prior is strongly shaped. Learning rate is low — the agent is resistant to change.

This trajectory mirrors Dirichlet parameter learning: early observations (with small concentration parameters) produce large updates; later observations (with large concentration parameters) produce small updates. The "10,000 hours" heuristic for expertise corresponds to the accumulation of sufficient concentration parameters to generate high-precision predictions.

2. Learning Plateaus and Phase Transitions

Embodied learning is not linear — it involves **plateaus** (periods of no apparent improvement) and **breakthroughs** (sudden competence jumps):

- **Plateaus** occur when the current generative model structure is being optimized locally — the parameters are being fine-tuned within the existing model architecture
- **Breakthroughs** occur when the model undergoes a **structural change** — a new level of hierarchy is added, a new latent variable is discovered, or a previously unsuspected relationship is recognized
- Example: A piano student may plateau for weeks while refining finger technique (parameter learning), then suddenly "break through" when they discover that phrasing is about breathing, not just notes (structural learning — a new variable enters the model)

3. Contemplative Practices as Precision Training

Meditation, breathwork, yoga, and body scanning are **systematic precision training** for the embodied generative model:

- **Mindfulness meditation**: Increases precision on present-moment interoceptive and exteroceptive observations while decreasing precision on self-referential narrative predictions (the "default mode network")
- **Yoga**: Increases precision on proprioceptive observations at extreme ranges of motion — expanding the domain over which the body-model makes accurate predictions

- **Breathwork**: Increases precision on respiratory interoception — training the agent to detect and modulate autonomic states through voluntary control of a normally automatic system

These practices don't add new information — they *change the precision structure* of the embodied generative model, making the agent more sensitive to signals that were previously below the attention threshold.

4. Trauma as Maladaptive Learning

Trauma can be understood as **maladaptive embodied learning** — a single high-impact experience (or repeated exposure) creates pathologically high-precision priors that distort ongoing inference:

- The traumatized agent's generative model overfits to the traumatic context → neutral stimuli generate threat prediction errors
- The body's interoceptive model becomes locked in a defensive state (hypervigilance, freeze, dissociation) → the learned prior overwhelms current evidence
- Somatic trauma therapies (EMDR, somatic experiencing, yoga therapy) work by gradually reducing the precision on traumatic priors — decoupling the somatic alarm signals from the current safe context

Applications

- **Rehabilitation after injury**: A patient relearning to walk after a stroke is updating their motor generative model to accommodate new neural pathways — the old B matrix (pre-stroke motor transitions) no longer holds, and the patient must build a new one through thousands of repetitions. Early sessions generate massive proprioceptive prediction errors; later sessions show progressive refinement.
- **Contemplative education**: Integrating mindfulness training into medical education systematically trains interoceptive precision in future clinicians — increasing their capacity for empathic resonance with patients and improving clinical decision-making through enhanced somatic awareness.

Conclusion

Somatic learning refines the embodied generative model through practice, contemplative training, and experiential engagement. The process involves parameter updating, structural phase transitions, precision training, and — when things go wrong — maladaptive learning that requires therapeutic intervention. The next module explores embodied communication — how somatic knowledge is shared between bodies.